

GRENADIER ISLAND, ON, Canada

WMO: 712810

Lat: 44.420N

Lon: 75.850W

Elev: 269

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-2.2	-16.2	2.3	-5.7	-10.5	3.2	-0.6	24.7	27.9	20.7	31.4	4.1	20	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	85.2	74.1	82.7	72.5	80.3	71.0	76.3	82.6	74.4	80.2	72.5	77.8	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	129.6	80.3	72.5	121.6	77.9	70.8	114.7	76.0	40.0	82.9	38.1	80.5	36.4	77.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	15.2	12.6	-14.0	89.3	6.5	2.8	-18.7	91.3	-22.6	93.0	-26.2	94.5	-31.0	96.6		
			-14.5	79.3	6.5	2.3	-19.1	80.9	-22.9	82.2	-26.6	83.5	-31.3	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	19.3	22.1	31.9	44.2	55.7	65.3	69.9	69.2	62.1	50.1	38.8	27.3	
	DBStd	19.54	12.95	10.68	10.34	7.97	6.70	5.85	4.58	5.03	6.61	7.58	8.56	10.01	
	HDD50	3671	952	782	566	205	21	0	0	0	2	94	345	704	
	HDD65	7210	1417	1202	1026	624	296	69	10	18	130	465	785	1168	
	CDD50	2382	0	0	5	32	198	458	618	596	366	97	11	1	
	CDD65	441	0	0	0	0	8	76	162	149	43	3	0	0	
	CDH74	2769	0	0	2	8	77	502	1015	901	252	12	0	0	
	CDH80	570	0	0	0	0	7	99	210	199	55	0	0	0	
Wind	WSAvg	5.3	6.0	5.6	5.7	5.9	4.9	4.2	4.4	4.2	4.9	5.4	6.2	6.1	
Precipitation	PrecAvg	38.7	2.9	2.4	2.6	3.1	3.3	3.2	3.6	3.1	3.6	4.1	3.6	3.3	
	PrecMax	48.3	6.1	4.2	4.8	5.7	5.7	6.9	7.8	4.6	6.8	7.2	5.7	5.6	
	PrecMin	30.8	1.0	1.0	1.1	0.4	1.3	0.7	1.3	0.9	1.9	0.7	1.7	1.5	
	PrecStd	4.2	1.3	0.7	1.0	1.4	1.3	1.5	1.5	1.2	1.2	1.8	1.0	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	48.4	67.7	73.6	80.5	87.0	87.6	88.2	86.0	75.6	62.9	53.9	
		MCWB	49.4	41.0	57.8	60.9	68.2	74.1	76.6	75.3	72.5	66.4	56.9	50.2	
	2%	DB	46.0	43.7	55.9	66.8	75.8	82.8	84.8	84.7	80.5	69.7	58.7	48.1	
		MCWB	44.4	40.3	48.6	53.7	62.9	72.3	74.9	73.4	71.0	63.0	54.2	45.5	
	5%	DB	40.3	40.2	50.6	61.4	72.1	79.6	82.3	82.0	76.2	65.5	55.1	44.4	
		MCWB	38.6	36.8	44.4	50.8	60.7	70.3	72.7	72.0	68.0	59.7	51.5	42.0	
	10%	DB	36.0	36.9	45.7	57.0	68.0	76.5	79.8	79.2	72.8	61.7	51.1	40.3	
		MCWB	34.4	34.5	40.7	48.3	58.8	67.9	71.4	70.0	65.8	56.9	47.7	38.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.8	43.9	58.0	62.6	71.0	77.0	79.3	78.3	74.6	68.7	58.4	51.9
			MCDB	51.1	45.7	66.2	70.8	77.4	84.8	85.5	84.8	82.3	72.8	60.8	54.1
2%		WB	44.2	40.6	50.0	56.6	66.2	73.9	76.5	75.9	71.8	63.6	55.2	46.1	
		MCDB	45.8	43.3	54.2	63.0	72.2	80.4	82.5	82.1	78.2	67.5	57.9	47.7	
5%		WB	38.7	37.4	45.3	52.8	63.2	71.4	74.7	73.8	69.5	60.6	52.1	42.1	
		MCDB	40.2	39.7	49.9	58.4	68.9	77.4	80.3	78.9	74.6	64.4	54.6	43.9	
10%		WB	34.5	34.7	41.3	49.3	60.4	69.3	72.7	71.9	67.5	57.5	48.0	38.4	
		MCDB	35.8	36.6	44.9	55.9	65.7	74.8	77.7	76.6	71.5	61.0	50.5	40.2	
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.2	16.5	17.4	18.0	17.9	17.1	17.5	17.2	14.6	12.6	12.3	
		MCDBR	15.3	16.4	22.5	24.9	23.9	21.5	19.4	20.3	21.1	19.6	18.0	14.8	
		MCWBR	14.6	13.4	15.2	14.1	12.9	12.3	11.1	11.0	12.0	13.1	14.0	13.0	
	5% WB	MCDBR	15.5	15.8	21.2	22.1	20.7	19.7	17.6	17.8	18.9	18.2	16.9	14.8	
		MCWBR	15.0	14.0	15.8	15.2	13.3	12.2	11.2	10.7	11.8	13.1	14.4	13.6	
Clear-Sky Solar Irradiance	taub	0.290	0.295	0.324	0.354	0.383	0.417	0.429	0.418	0.369	0.349	0.310	0.285		
	taud	2.374	2.362	2.417	2.394	2.357	2.278	2.251	2.282	2.407	2.456	2.509	2.455		
	Ebn at Noon	263	284	290	288	281	271	266	265	271	261	257	256		
	Edh at Noon	24	29	32	36	38	42	42	40	32	27	22	20		
All-Sky Solar Radiation	RadAvg	426	706	1126	1399	1678	1794	1837	1618	1288	765	480	328		
	RadStd	42	71	92	169	173	144	158	75	109	73	53	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page